HDOC - 67th day

GCD USING EUCLID'S ALGORITHM

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Problem Statement

Write a recursive function to find the GCD of two positive integers using Euclid's algorithm.

The program should read two positive integers, recursively compute their greatest common divisor using GCD(a, b) = GCD(b, a % b), and display the result.

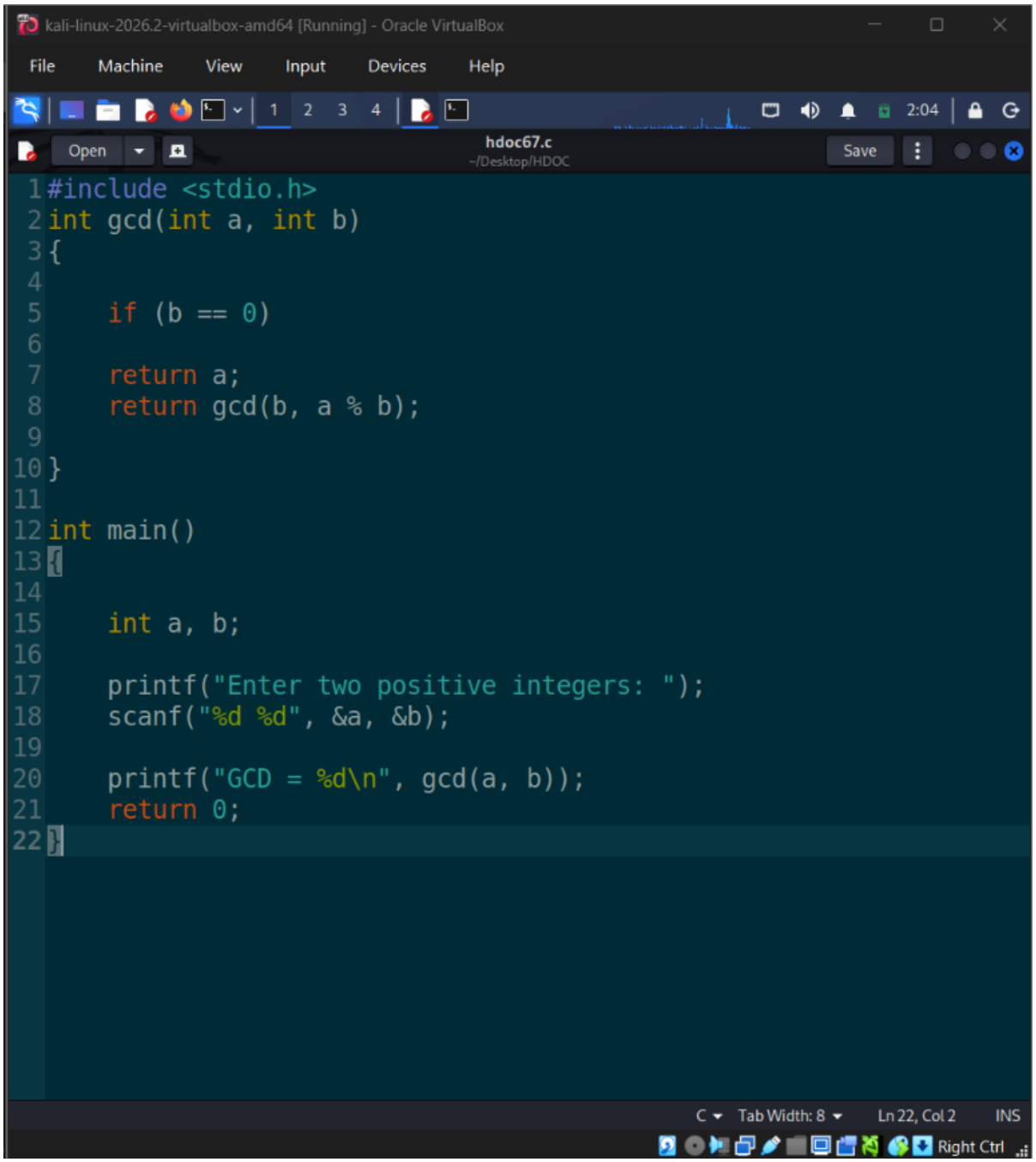
Algorithm

1. Start.
2. Read two positive integers a and b.
3. Call the recursive function gcd(a, b).
4. If b = 0, return a as the GCD.
5. Otherwise, recursively call gcd(b, a % b).
6. Continue until the second number becomes 0.
7. Display the GCD returned by the recursive function.
8. Stop.

Test Case Table

S. No.	First Integer	Second Integer	Expected Output	Actual Output	Result
1	48	18	GCD = 6	GCD = 6	Pass
2	100	25	GCD = 25	GCD = 25	Pass
3	17	13	GCD = 1	GCD = 1	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
hdoc67.c
~/Desktop/HDOC
Save
#include <stdio.h>
int gcd(int a, int b)
{
if (b == 0)
return a;
return gcd(b, a % b);
}
int main()
{
int a, b;
printf("Enter two positive integers: ");
scanf("%d %d", &a, &b);
printf("GCD = %d\n", gcd(a, b));
return 0;
}
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Compilation and Execution

Compilation command: gcc hdoc67.c -o hdoc67



Execution command: ./hdoc67

Execution Screenshot:

